

Unitech RFID Command Set

Version: V1.02

Note: [New/Updated functions in blue](#)

Applied to FW: RM300P V0.1.0.14 or later

1. Command Format

1.1. Host Command Formats

<STX><LENGTH><ID><CHECKSUM><ETX>

<STX><LENGTH><ID><DATA><CHECKSUM><ETX>

STX	1 Byte	; 0x02 (Start of the command)
LENGTH	2 Byte	; Length of command in bytes, excluding checksum and STX/ETX
ID	2 Byte	; Identifier for the command
DATA	N Byte	; Command data
CHECKSUM	2 Byte	; 2's complement sum of message (LENGTH+ID+DATA), excluding Checksum and STX/ETX
ETX	1 Byte	; 0x03 (End of the command)

1.2. RFID Response Formats

<STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

<STX><LENGTH><ID><STATUS><DATA><CHECKSUM><ETX>

STX	1 Byte	; 0x02 (Start of the command)
LENGTH	2 Byte	; Length of command in bytes, excluding checksum and STX/ETX
ID	2 Byte	; Identifier for the command
STATUS	1 Byte	; Status; 0 for Success
DATA	N Byte	; Response data
CHECKSUM	2 Byte	; 2's complement sum of message (LENGTH+ID+DATA), excluding Checksum and STX/ETX
ETX	1 Byte	; 0x03 (End of the command)

Note:

STATUS	=	0x00	; Success
		0x01	; Busy
		0x02	; Invalid parameters
		0x03	; Not Supported
		0x81	; Format Error (ID:0xFFFF) – incorrect Length
		0x82	; Format Error (ID:0xFFFF) – incorrect CRC
		0x83	; Format Error (ID:0xFFFF) – incorrect End-Of-Packet
		0x84	; Format Error (ID:0xFFFF) – packet timeout

0xE0	; Other Error
0xE1	; Target Not Found

1.3. RFID Data Send Format

<STX><LENGTH><ID><DATA><CHECKSUM><ETX>

STX	1 Byte	; 0x02 (Start of the command)
LENGTH	2 Byte	; Length of command in bytes, excluding checksum and STX/ETX
ID	2 Byte	; 0x80 for the command
DATA	N Byte	; Data sent to the host
CHECKSUM	2 Byte	; 2's complement sum of message (LENGTH+ID+DATA), excluding ; Checksum and STX/ETX
ETX	1 Byte	; 0x03 (End of the command)

2. General Commands

2.1. Read Firmware Version Command (ID = 0x21)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <DATA> <CHECKSUM> <ETX>

Note: DATA = <MAJOR> <MINOR> <BUILD> <REVISION>

Example:

Host ⇒ <0x02> <0x00 0x04> <0x00 0x21> <0xFF 0xDB> <0x03>

RFID ⇒ <0x02> <0x00 0x9> <0x00 0x21> <0x00> <0x00 0x00 0x00 0x01> <0xFF 0xD5> <0x03>
; Version 0.0.0.1

2.2. Read Model Name Command (ID = 0x22)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <DATA> <CHECKSUM> <ETX>

Example:

Host ⇒ <0x02> <0x00 0x04> <0x00 0x22> <0xFF 0xDA> <0x03>

RFID ⇒ <0x02> <0x00 0x0B> <0x00 0x22> <0x00> <RM300P> <0xFE 0x51> <0x03>

2.3. Read Serial Number Command (ID = 0x24)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <DATA> <CHECKSUM> <ETX>

Example:

Host ⇒ <0x02> <0x00 0x04> <0x00 0x24> <0xFF 0xD8> <0x03>

RFID ⇒ <0x02> <0x00 0x11> <0x00 0x24> <0x00> <123456789012> <0xFD 0x5B> <0x03>

2.4. Read SKU ID Command (ID = 0x25)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <DATA> <CHECKSUM> <ETX>

Example:

Host ⇒ <0x02> <0x00 0x04> <0x00 0x25> <0xFF 0xD7> <0x03>

RFID ⇒ <0x02> <0x00 0x06> <0x00 0x25> <0x00> <0x00> <0xFF 0xD5> <0x03>

2.5. Read Ambient Temperature Command (ID = 0x28)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <TEMPDATA> <CHECKSUM> <ETX>

Note:

TEMPDATA 2 Byte ; Temperature Data in tenth of Celsius.

Example:

Host ⇒ <0x02><0x00 0x04><0x00 0x28><0xFF 0xD4><0x03>

RFID ⇒ <0x02><0x00 0x07><0x00 0x28><0x00><0x01 0xC0><0xFF 0x10><0x03>

2.6. Read PA Temperature Command (ID = 0x29)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><TEMPDATA><CHECKSUM><ETX>

Note:

TEMPDATA 2 Byte ; Temperature Data in tenth of Celsius.

Example:

Host ⇒ <0x02><0x00 0x04><0x00 0x29><0xFF 0xD3><0x03>

RFID ⇒ <0x02><0x00 0x07><0x00 0x29><0x00><0x01 0xC0><0xFF 0x0F><0x03>

2.7. Reset Device Command (ID = 0x2F)

Host ⇒ <STX><LENGTH><ID><FLAGS><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

FLAGS 1 Byte ; 0x00 Reboot only

 ; 0x01 Reboot & Factory Reset.

Example:

Host ⇒ <0x02><0x00 0x05><0x00 0x2F><0x00><0xFF 0xCC><0x03>

RFID ⇒ <0x02><0x00 0x05><0x00 0x2F><0x00><0xFF 0xCC><0x03>

3. RFID Operation

3.1. Tag Inventory Operation Command (ID = 0x80)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Example:

Host ⇒ <0x02> <0x00 0x04> <0x00 0x80> <0xFF 0x7C> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x00 0x80> <0x00> <0xFF 0x7B> <0x03>

; Tag Data Reply – Inventory Data. See Ch4.1

RFID ⇒ <0x02> <0x00 0x15> <0x00 0xC0> <0x01> <0xE9 0x62> <0x34 0x00>

<0xAA 0xAA 0x34 0x12 0xDC 0x03 0x01 0x17 0x16 0x10 0xB9 0x88> <0xF9 0xB3> <0x03>

3.2. Cancel Operation Command (ID = 0x81)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Example:

Host ⇒ <0x02> <0x00 0x04> <0x00 0x81> <0xFF 0x7B> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x00 0x81> <0x00> <0xFF 0x7A> <0x03>

3.3. Tag Read Operation Command (ID = 0x82)

Host ⇒ <STX> <LENGTH> <ID> <ACCPW> <MBANK> <MADDR> <MLEN> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

ACCPW	4 Byte	; Access password
MBANK	1 Byte	; Memory Bank i.e. RESV/EPC/TID/USER
MADDR	2 Byte	; Memory Start Address (in words)
MLEN	1 Byte	; Memory Length (in words)

Example:

Host ⇒ <0x02> <0x00 0x0C> <0x00 0x82> <0x12 0x34 0x56 0x78> <0x01>

<0x00 0x02> <0x01> <0xFE 0x5A> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x00 0x82> <0x00> <0xFF 0x79> <0x03>

; Tag Data Reply – Tag Access Data. See Ch4.2

RFID ⇒ <0x02> <0x00 0x09> <0x00 0xC8> <0x00 0x82> <0x00> <0xAA 0xAA>

<0xFD 0x59> <0x03>

3.4. Tag Write Operation Command (ID = 0x83)

Host ⇒ <STX><LENGTH><ID><ACCPW><MBANK><MADDR><MLEN><MDATA>
<CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ACCPW	4 Byte	; Access password
MBANK	1 Byte	; Memory Bank i.e. RESV/EPC/TID/USER
MADDR	2 Byte	; Memory Start Address (in words)
MLEN	1 Byte	; Memory Length (in words)
MDATA	2N Byte	; Memory Data (in words)

Example:

Host ⇒ <0x02><0x00 0x0E><0x00 0x83><0x12 0x34 0x56 0x78>
<0x01><0x00 0x02><0x01><0xBB 0xBB><0xFC 0xE1><0x03>

RFID ⇒ <0x02><0x00 0x05><0x00 0x83><0x00><0xFF 0x78><0x03>

; Tag Data Reply – Tag Access Data. See Ch4.2

RFID ⇒ <0x02><0x00 0x07><0x00 0xC8><0x00 0x83><0x00><0xFE 0xAE><0x03>

3.5. Tag Lock Operation Command (ID = 0x84)

Host ⇒ <STX><LENGTH><ID><ACCPW><LDATA><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ACCPW	4 Byte	; Access password
LDATA	3 Byte	; Lock Command Data. (Refer to ISO 18000-6C)

Example:

Host ⇒ <0x02><0x00 0x0B><0x00 0x84><0x12 0x34 0x56 0x78><0x00 0x80 0x20>
<0xFD 0xBD><0x03>

RFID ⇒ <0x02><0x00 0x05><0x00 0x84><0x00><0xFF 0x77><0x03>

; Tag Data Reply – Tag Access Data. See Ch4.2

RFID ⇒ <0x02><0x00 0x07><0x00 0xC8><0x00 0x84><0x00><0xFE 0xAD><0x03>

3.6. Tag Kill Operation Command (ID = 0x85)

Host ⇒ <STX><LENGTH><ID><KILLPW><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

KILLPW	4 Byte	; Kill password
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Example:

Host ⇒ <0x02><0x00 0x08><0x00 0x85><0xFF 0xFF 0xFF 0xFF><0xFB 0x77><0x03>

RFID ⇒ <0x02><0x00 0x05><0x00 0x85><0x00><0xFF 0x76><0x03>

; Tag Data Reply – Tag Access Data. See Ch4.2

RFID ⇒ <0x02><0x00 0x07><0x00 0xC8><0x00 0x85><0x00><0xFE 0xAC><0x03>

3.7. Tag BlockWrite Operation Command (ID = 0x86)

Host ⇒ <STX><LENGTH><ID><ACCPW><MBANK><MADDR><MLEN><MDATA>
<CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ACCPW	4 Byte	; Access password
MBANK	1 Byte	; Memory Bank i.e. RESV/EPC/TID/USER
MADDR	2 Byte	; Memory Start Address (in words)
MLEN	1 Byte	; Memory Length (in words)
MDATA	2N Byte	; Memory Data (in words)

Example:

Host ⇒ <0x02><0x00 0x14><0x00 0x86><0x12 0x34 0x56 0x78><0x03>
<0x00 0x00><0x04><0xAA 0xAA 0xBB 0xBB 0xCC 0xCC 0xDD 0xDD>
<0xF8 0x2F><0x03>

RFID ⇒ <0x02><0x00 0x05><0x00 0x86><0x00><0xFF 0x75><0x03>

; Tag Data Reply – Tag Access Data. See Ch4.2

RFID ⇒ <0x02><0x00 0x07><0x00 0xC8><0x00 0x86><0x00><0xFE 0xAB><0x03>

3.8. Tag BlockPermalock Operation Command (ID = 0x87)

Host ⇒ <STX><LENGTH><ID><ACCPW><READLOCK><MBANK><MADDR><MLEN><MSAK>
<CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ACCPW	4 Byte	; Access password
READLOCK	1 Byte	; Read or Permalock. i.e. READ/LOCK
MBANK	1 Byte	; Memory Bank i.e. RESV/EPC/TID/USER
MADDR	2 Byte	; Memory Start Address (in 16 blocks)
MLEN	1 Byte	; Memory Range (in 16 blocks)
MASK	2N Byte	; Retain current or Assert Permalock

Example:

Host ⇒ <0x02><0x00 0x0F><0x00 0x87><0x00 0x00 0x00 0x00><0x01><0x03>
<0x00 0x00><0x01><0x80 0x00><0xFE 0xE5><0x03>

RFID ⇒ <0x02><0x00 0x05><0x00 0x87><0x00><0xFF 0x74><0x03>

; Tag Data Reply – Tag Access Data. See Ch4.2

RFID ⇒ <0x02><0x00 0x07><0x00 0xC8><0x00 0x87><0x00><0xFE 0xAA><0x03>

3.9. Tag Authenticate Operation Command (ID = 0x88)

Host ⇒ <STX><LENGTH><ID><ACCPW><SENREP><INCREPLEN><CSI><LEN><MSG>
<CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ACCPW	4 Byte	; Access password
SENREP	1 Byte	; Store or Send
INCREPLEN	1 Byte	; Includes Length in reply.
CSI	1 Byte	; Cryptographic suite
LEN	2 Byte	; Length of Message (in bits)
MSG	N Byte	; Message (Depends on CSI)
EXPREPLEN	2 Byte	; Expected Length of Response (in bits)

Example:

Host ⇒ <0x02><0x00 0x15><0x00 0x88><0x00 0x00 0x00 0x00><0x01><0x01><0x01>
<0x00 0x30><0x05 0x02 0x03 0x04 0x05 0x06><0x00 0x80><0xFE 0x97><0x03>

RFID ⇒ <0x02><0x00 0x05><0x00 0x88><0x00><0xFF 0x73><0x03>

; Tag Data Reply – Tag Access Data. See Ch4.2

; LenRep: 0x80; Parity: 0x01; Response: 11A210614C45FD00E4896532AFB40030

RFID ⇒ <0x02><0x00 0x19><0x00 0xC8><0x00 0x88><0x00><0x01 0x01 0x11 0xA2
0x10 0x61 0x4C 0x45 0xFD 0x00 0xE4 0x89 0x65 0x32 0xAF 0xB4 0x00 0x30>
<0xF8 0x4C><0x03>

3.10. Tag Select Operation Command (ID = 0x8A)

Host ⇒ <STX><LENGTH><ID><MODE><SELTARGET><ACTION><TRUNC>
<MBANK><MADDR><MLEN><MDATA><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

MODE	1 Byte	; Mode; Clear:0, Add:1, Clear_Add:2, AddOnce:3
SELTARGET	1 Byte	; Select Target; 0:S0, 1:S1, 2:S2, 3:S3, 4:SL
ACTION	1 Byte	; Select Action;
TRUNC	1 Byte	; Enable Truncation.
MBANK	1 Byte	; Memory Bank i.e. RESV/EPC/TID/USER
MADDR	2 Byte	; Memory Start Address (in bits)
MLEN	1 Byte	; Memory Length (in bits)
MDATA	N Byte	; Memory Data-Pattern

Example:

Host ⇒ <0x02> <0x00 0x0E> <0x00 0x8A> <0x02> <0x04> <0x00> <0x00> <0x01>
 <0x00 0x20> <0x10> <0xAA 0xAA> <0xFD 0xDD> <0x03>
 RFID ⇒ <0x02> <0x00 0x05> <0x00 0x8A> <0x00> <0xFF 0x71> <0x03>

3.11. Tag Inventory with User-Specified Data Operation Command (ID = 0x8B)

Host ⇒ <STX> <LENGTH> <ID> <ACCPW> <MBANK> <MADDR> <MLEN> <CHECKSUM> <ETX>
 RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

ACCPW	4 Byte	; Access password
MBANK	1 Byte	; Memory Bank i.e. RESV/EPC/TID/USER
MADDR	2 Byte	; Memory Start Address (in words)
MLEN	1 Byte	; Memory Length (in words)

Example:

Host ⇒ <0x02> <0x00 0x0C> <0x00 0x8B> <0x12 0x34 0x56 0x78> <0x01>
 <0x00 0x02> <0x01> <0xFE 0x4D> <0x03>
 RFID ⇒ <0x02> <0x00 0x05> <0x00 0x8B> <0x00> <0xFF 0x70> <0x03>

; Tag Data Reply – Inventory Data. See Ch4.1

RFID ⇒ <0x02> <0x00 0x16> <0x00 0xC2> <0x01> <0xE9 0x62> <0x34 0x00>
 <0xAA 0xAA 0x34 0x12 0xDC 0x03 0x01 0x17 0x16 0x10 0xB9 0x88>
 <0x1> <0xF9 0xAF> <0x03>

3.12. Tag MarginRead Operation Command (ID = 0x8C)

Host ⇒ <STX> <LENGTH> <ID> <ACCPW> <MBANK> <MADDR> <MLEN> <MDATA>
 <CHECKSUM> <ETX>
 RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

ACCPW	4 Byte	; Access password
MBANK	1 Byte	; Memory Bank i.e. RESV/EPC/TID/USER
MADDR	2 Byte	; Memory Start Address (in bits)
MLEN	1 Byte	; Memory Length (in bits)
MDATA	N Byte	; Memory Data-Pattern

Example:

Host ⇒ <0x02> <0x00 0x0D> <0x00 0x8C> <0x00 0x00 0x00 0x00> <0x03> <0x00 0x00>
 <0x08> <0x05> <0xFF 0x07> <0x03>
 RFID ⇒ <0x02> <0x00 0x05> <0x00 0x8C> <0x00> <0xFF 0x6F> <0x03>

; Tag Data Reply – Inventory Data. See Ch4.1

; Tag Data Reply – Success

RFID ⇒ <0x02><0x00 0x07><0x00 0xC8><0x00 0x8C><0x00><0xFE 0xA5><0x03>

; *Tag Data Reply – Fail*

RFID ⇒ <0x02><0x00 0x07><0x00 0xC8><0x00 0x8C><0x80><0xFE 0x25><0x03>

; *Tag Data Reply – No Tag Found*

RFID ⇒ <0x02><0x00 0x07><0x00 0xC8><0x00 0x8C><0x01><0xFE 0xA4><0x03>

; *Tag Data Reply – Tag No Reply*

RFID ⇒ <0x02><0x00 0x07><0x00 0xC8><0x00 0x8C><0x03><0xFE 0xA2><0x03>

4. RFID Data Reply

4.1. RFID Data Reply – Inventory Data (ID = 0xC0)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <ANTID><RSSI><PC><EPC>

ANTID	1 Byte	; Antenna ID; 1~4
RSSI	2 Byte	; RSSI value
PC	2 Byte	; Protocol Control of EPC bank
EPC	N Byte	;EPC value of EPC bank

Example:

RFID ⇒ <0x02><0x00 0x15><0x00 0xC0><0x01><0xE9 0x62><0x34 0x00>
<0xAA 0xAA 0x34 0x12 0xDC 0x03 0x01 0x17 0x16 0x10 0xB9 0x88><0xF9 0xB3><0x03>

4.2. RFID Data Reply – Inventory Data with TID (ID = 0xC1)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <ANTID><RSSI><PC><EPC><TID>

ANTID	1 Byte	; Antenna ID; 1~4
RSSI	2 Byte	; RSSI value
PC	2 Byte	; Protocol Control of EPC bank
EPC	N Byte	;EPC value of EPC bank
TID	N Byte	;TID value of TID bank

Example:

RFID ⇒ <0x02><0x00 0x21><0x00 0xC1><0x01><0xF1 0x2A><0x34 0x00>
<0xBB 0xBB 0x34 0x12 0xDC 0x03 0x01 0x17 0x16 0x10 0xB9 0x88>
<0xE2 0x00 0x34 0x12 0x01 0x38 0x01 0x00 0x06 0x7A 0x8E 0x7F>
<0xF6 0xC5><0x03>

4.3. RFID Data Reply – Inventory Data with User-Specified Data (ID = 0xC2)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <ANTID><RSSI><PC><EPC><TID>

ANTID	1 Byte	; Antenna ID; 1~4
RSSI	2 Byte	; RSSI value
PC	2 Byte	; Protocol Control of EPC bank
EPC	N Byte	;EPC value of EPC bank
DATA	N Byte	;User-Specified Data

Example:

RFID ⇒ <0x02><0x00 0x16><0x00 0xC2><0x01><0xE9 0x62><0x34 0x00>
<0xAA 0xAA 0x34 0x12 0xDC 0x03 0x01 0x17 0x16 0x10 0xB9 0x88>
<0x01><0xF9 0xAF><0x03>

4.4. RFID Data Reply – Inventory Data/Phase (ID = 0xB0)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <ANTID><RSSI><PC><EPC>

ANTID	1 Byte	; Antenna ID; 1~4
RSSI	2 Byte	; RSSI value
PHASEBEGIN	2 Byte	; Phase Begin in 0.087 degrees
PHASEEND	2 Byte	; Phase End in 0.087 degrees
FREQ	4 Byte	; Frequency in kHz; e.g. 902750
PC	2 Byte	; Protocol Control of EPC bank
EPC	N Byte	;EPC value of EPC bank

Example:

RFID ⇒ <0x02><0x00 0x1D><0x00 0xB0><0x01><0xE7 0xFC>
<0x02 0xF5><0x02 0xE8><0x00 0x0D 0xDD 0xCE><0x34 0x04>
<0x00 0x00 0x00 0x00 0x00 0x00 0x02 0x02 0x10 0x83 0x14 0x16><0xF8 0xBD><0x03>

4.5. RFID Data Reply – Inventory Data/Phase with TID (ID = 0xB1)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <ANTID><RSSI><PC><EPC><TID>

ANTID	1 Byte	; Antenna ID; 1~4
RSSI	2 Byte	; RSSI value
PHASEBEGIN	2 Byte	; Phase Begin in 0.087 degrees
PHASEEND	2 Byte	; Phase End in 0.087 degrees
FREQ	4 Byte	; Frequency in kHz; e.g. 902750
PC	2 Byte	; Protocol Control of EPC bank
EPC	N Byte	;EPC value of EPC bank
TID	N Byte	;TID value of TID bank

Example:

RFID ⇒ <0x02><0x00 0x29><0x00 0xB1><0x01><0xF1 0x92>
<0x01 0x1C><0x01 0x0F><0x00 0x0D 0xCA 0x46><0x34 0x00>
<0xE2 0x00 0x00 0x1B 0x81 0x09 0x01 0x17 0x8B 0xCC 0xB1 0x47>
<0xE2 0x00 0x34 0x12 0x01 0x3A 0x01 0x00 0x06 0x7A 0x86 0x3E>
<0xF5 0x9E><0x03>

4.6. RFID Data Reply – Inventory Data/Phase with User-Specified Data (ID = 0xB2)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <ANTID><RSSI><PC><EPC><TID>

ANTID	1 Byte	; Antenna ID; 1~4
RSSI	2 Byte	; RSSI value
PHASEBEGIN	2 Byte	; Phase Begin in 0.087 degrees

PHASEEND	2 Byte	; Phase End in 0.087 degrees
FREQ	4 Byte	; Frequency in kHz; e.g. 902750
PC	2 Byte	; Protocol Control of EPC bank
EPC	N Byte	;EPC value of EPC bank
DATA	N Byte	;User-Specified Data

Example:

RFID ⇒ <0x02><0x00 0x1F><0x00 0xB2><0x01><0xE8 0xE3>
 <0x04 0x29><0x04 0x02><0x00 0x0D 0xD9 0xE6><0x34 0x00>
 <0xE2 0x00 0x34 0x12 0xDC 0x03 0x01 0x17 0x75 0x16 0x60 0x13>
 <0xE2 0x00><0xF7 0x31><0x03>

4.7. RFID Data Reply – Inventory Data with Customized Format (ID = 0xA0)

RFID ⇒ <STX><LENGTH><ID><PARAMETER1>...<PARAMETERn><CHECKSUM><ETX>

Note:

PARAMETER Format:

PLEN	2 Byte	; Length of PLEN, PID and PDATA in byte
PID	1 Byte	; Parameter ID
PDATA	N Byte	

PID	= 0x00	; EPC
	= 0x01	; PC
	= 0x02	; RSSI
	= 0x03	; Antenna ID
	= 0x04	; TID
	= 0x05	; User specified memory
	= 0x06	; Phase
	= 0x07	; Frequency
	= 0x08	; Timestamp

PDATA Format:

Phase:

Phase Begin	2 Byte	; Phase Begin in 0.087 degrees
Phase End	2 Byte	; Phase End in 0.087 degrees

Timestamp:

YEAR	1 Byte	; Year (From 2000, e.g. 1 for 2001)
MON	1 Byte	; Month
DAY	1 Byte	; Day
HOU	1 Byte	; Hour
MIN	1 Byte	; Minute
SEC	1 Byte	; Second

Example:

RFID ⇒ <0x02><0x00 0x1C><0x00 0xA0><0x00 0x0F><0x00>
 <0xE2 0x00 0x34 0x12 0xDC 0x03 0x01 0x11 0x77 0x43 0x31 0x43 0x80>
 <0x00 0x09><0x08><0x19 0x03 0x15 0x0E 0x21 0x39>
 <0xFB 0x79><0x03>

4.8. RFID Data Reply – Inventory Status (ID = 0xC7)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <INVSTATUS>

INVSTATUS 1 Byte ; Inventory Status; 0: Inventory Done.

Example:

RFID ⇒ <0x02><0x00 0x05><0x00 0xC7><0x00><0xFF 0x34><0x03>

4.9. RFID Data Reply – Tag Access Data (ID = 0xC8)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <OPID><TAERRCODE><MDATA>

OPID 2 Byte ; Operation ID

TAERRCODE 1 Byte ; Tag Access Error Code

MDATA N Byte ; OpID=0x82 : Memory Data (For OpID:0x82)
 ; OpID=0x87 : Mask (if READLOCK=0)
 ; OpID=0x88 : Len (if INCREPLEN=1) + Response
 ; Len: 15 Bits for LenRep + 1 Bit for Parity

OPID = 0x82 ; Tag Read Data

= 0x83 ; Tag Write Data

= 0x84 ; Tag Lock

= 0x85 ; Tag Kill

= 0x86 ; Tag Block Write

= 0x87 ; Tag Block PermaLock

= 0x88 ; Tag Authenticate

TAERRCODE = 0x00 ; Success

= 0x01 ; No-Tag found

= 0x02 ; Tag Lost

= 0x03 ; Tag Action Not Reply

= 0x04 ; Invalid Password

= 0x05 ; Zero Kill Password

= 0x80~0x8F ; Tag-Reply Error (Gen2 Spec)

Example:

RFID ⇒ <0x02><0x00 0x09><0x00 0xC8><0x00 0x82><0x00><0xAA 0xAA>

<0xFD 0x59> <0x03>

4.10. RFID Data Reply– Operation Error (ID = 0xD0)

RFID ⇒ <STX> <LENGTH> <ID> <DATA> <CHECKSUM> <ETX>

Note: <DATA> = <ANTID> <OPERRCODE>

ANTID 1 Byte ; Antenna ID; 1~4

OPERRCODE 1 Byte ; Operation Error Code

OPERRCODE = 0x00 ; Success

= 0x01~0x11 ; Ex10 Operation Error

= 0xD0 ; Ex10 Initialized Error

= 0xD1 ; No Antenna Enabled Error

= 0xD2 ; Antenna Disconnected Error

= 0xD3 ; Incorrect DwellTime/RoundCount Error

Example:

RFID ⇒ <0x02> <0x00 0x06> <0x00 0xD0> <0x04> <0x0A> <0xFF 0x1C> <0x03>

4.11. RFID Data Reply– System Error (ID = 0xD1)

RFID ⇒ <STX> <LENGTH> <ID> <DATA> <CHECKSUM> <ETX>

Note: <DATA> = <SYSERRCODE>

SYSERRCODE 1 Byte ; System Error Code

SYSERRCODE = 0x01 ; Ambient Temperature Too High

= 0x02 ; PA Temperature Too High

Example:

RFID ⇒ <0x02> <0x00 0x05> <0x00 0xD1> <0x01> <0xFF 0x29> <0x03>

5. RFID Configuration

5.1. Get Region Command (ID = 0x100)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><DATA><CHECKSUM><ETX>

Note: DATA = 0 ; FCC
= 1 ; ETSI
= 2 ; JAPAN
= 3 ; TAIWAN
= 4 ; CHINA
= 5 ; HONG-KONG

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x00><0xFF 0xFB><0x03>

RFID ⇒ <0x02><0x00 0x06><0x01 0x00><0x00><0x00><0xFF 0xF9><0x03> ; FCC

5.2. Set Region Command (ID = 0x101)

Host ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note: DATA = 0 ; FCC
= 1 ; ETSI
= 2 ; JAPAN
= 3 ; TAIWAN
= 4 ; CHINA
= 5 ; HONG-KONG

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0x01><0x01><0xFF 0xF8><0x03> ; ETSI

RFID ⇒ <0x02><0x00 0x05><0x01 0x01><0x00><0xFF 0xF9><0x03>

5.3. Get Antenna Settings Command (ID = 0x102)

Host ⇒ <STX><LENGTH><ID><ANTID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><ANTID><DWELL><PWR><CHECKSUM><ETX>

Note:

ANTID 1 Byte ; Antenna ID; 1~4
DWELL 2 Byte ; Dwell Time in ms
PWR 2 Byte ; Antenna Power in dBm. (Hundredths of dBm)

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0x02><0x01><0xFF 0xF7><0x03>

RFID ⇒ <0x02><0x00 0x0A><0x01 0x02><0x00><0x01><0x03 0xE8><0x0B 0xB8>
<0xFE 0x44><0x03>

5.4. Set Antenna Settings Command (ID = 0x103)

Host ⇒ <STX><LENGTH><ID><ANTID><DWELL><PWR><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ANTID	1 Byte	; Antenna ID; 1~4
DWELL	2 Byte	; Dwell Time in ms
PWR	2 Byte	; Antenna Power in cdBm. (Hundredths of dBm)

Example:

Host ⇒ <0x02><0x00 0x09><0x01 0x03><0x01><0x03 0xE8><0x0B 0xB8><0xFE 0x44><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x03><0x00><0xFF 0xF7><0x03>

5.5. Get Antenna State Command (ID = 0x104)

Host ⇒ <STX><LENGTH><ID><ANTID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><ANTID><ANTSTATE><CHECKSUM><ETX>

Note:

ANTID	1 Byte	; Antenna ID; 1~4
ANTSTATE	1 Byte	; 0: Disable; 1: Enable

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0x04><0x01><0xFF 0xF5><0x03>

RFID ⇒ <0x02><0x00 0x07><0x01 0x04><0x00><0x01><0x01><0xFF 0xF2><0x03>

5.6. Set Antenna State Command (ID = 0x105)

Host ⇒ <STX><LENGTH><ID><ANTID><ANTSTATE><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ANTID	1 Byte	; Antenna ID; 1~4
ANTSTATE	1 Byte	; 0: Disable; 1: Enable

Example:

Host ⇒ <0x02><0x00 0x06><0x01 0x05><0x01><0x00><0xFF 0xF3><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x05><0x00><0xFF 0xF5><0x03>

5.7. Get RF Mode Command (ID = 0x106)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><RFMODE><CHECKSUM><ETX>

Note:

RFMODE	2 Byte	; RF Mode. e.g. 103
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Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x06><0xFF 0xF5><0x03>
 RFID ⇒ <0x02><0x00 0x07><0x01 0x06><0x00><0x00 0x94><0xFF 0x5E><0x03>

5.8. Set RF Mode Command (ID = 0x107)

Host ⇒ <STX><LENGTH><ID><RFMODE><CHECKSUM><ETX>
 RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

RFMODE 2 Byte ; RF Mode. e.g. 103

Example:

Host ⇒ <0x02><0x00 0x06><0x01 0x07><0x00 0x94><0xFF 0x5E><0x03>
 RFID ⇒ <0x02><0x00 0x05><0x01 0x07><0x00><0xFF 0xF3><0x03>

5.9. Get Gen2 Algorithm Command (ID = 0x108)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>
 RFID ⇒ <STX><LENGTH><ID><STATUS><ALG><INITQ><QRANGE><DUALTARGET>
 <CHECKSUM><ETX>

Note:

ALG 1 Byte ; Algorithm. i.e. 0: Fix Q ; 1: Dynamic Q
 INITQ 1 Byte ; Q Value or Start Q.
 QRANGE 1 Byte ; Bit 0-3: Min Q; Bit 4-7 : Max Q ;
 DUALTARGET 1 Byte ; Dual Target;

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x08><0xFF 0xF3><0x03>
 RFID ⇒ <0x02><0x00 0x09><0x01 0x08><0x00><0x00><0x04><0x44><0x01>
 <0xFF 0xA5><0x03>

5.10. Set Gen2 Algorithm Command (ID = 0x109)

Host ⇒ <STX><LENGTH><ID><ALG><INITQ><QRANGE><DUALTARGET>
 <CHECKSUM><ETX>
 RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ALG 1 Byte ; Algorithm. i.e. 0: Fix Q ; 1: Dynamic Q
 INITQ 1 Byte ; Q Value or Start Q.
 QRANGE 1 Byte ; Bit 0-3: Min Q; Bit 4-7 : Max Q ;
 DUALTARGET 1 Byte ; Dual Target;

Example:

Ex1. Set Fixed Q, Q Value = 4, Min Q = 4, Max Q = 4, Dual Target = Enable

Host ⇒ <0x02><0x00 0x08><0x01 0x09><0x00><0x04><0x44><0x01><0xFF 0xA5><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x09><0x00><0xFF 0xF1><0x03>

Ex2. Set Dynamic Q, Initial Q = 8, Min Q = 0, Max Q = 15, Dual Target = Enable

Host ⇒ <0x02><0x00 0x08><0x01 0x09><0x01><0x08><0xF0><0x01><0xFE 0xF4><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x09><0x00><0xFF 0xF1><0x03>

5.11. Get Gen2 Query Group Command (ID = 0x10A)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><SELECT><SESSION><TARGET>
<CHECKSUM><ETX>

Note:

SELECT	1 Byte	; Select Mode; 1: All; 2 De-Selected; 3: Selected
SESSION	1 Byte	; Session: S0~S3
TARGET	1 Byte	; 0: Target A; 1: Target B;

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x0A><0xFF 0xF1><0x03>

RFID ⇒ <0x02><0x00 0x08><0x01 0x0A><0x00><0x01><0x00><0x00><0xFF 0xEC><0x03>

5.12. Set Gen2 Query Group Command (ID = 0x10B)

Host ⇒ <STX><LENGTH><ID><SELECT><SESSION><TARGET>
<CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

SELECT	1 Byte	; Select Mode; 1: All; 2 De-Selected; 3: Selected
SESSION	1 Byte	; Session: S0~S3
TARGET	1 Byte	; 0: Target A; 1: Target B;

Example:

Host ⇒ <0x02><0x00 0x07><0x01 0x0B><0x01><0x00><0x00><0xFF 0xEC><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x0B><0x00><0xFF 0xEF><0x03>

5.13. Get Bi-Static Antenna Setting Command (ID = 0x10C)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><BISTATICEN><CHECKSUM><ETX>

Note:

BISTATICEN	1 Byte	; Bi-Static Enable/Disable; 0: Disable; 1: Enable
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Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x0C><0xFF 0xEF><0x03>

RFID ⇒ <0x02><0x00 0x06><0x01 0x0C><0x00><0x00><0xFF 0xED><0x03>

5.14. Set Bi-Static Antenna Settings Command (ID = 0x10D)

Host ⇒ <STX> <LENGTH> <ID> <BISTATICEN> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

BISTATICEN 1 Byte ; Bi-Static Enable/Disable; 0: Disable; 1: Enable

Example:

Host ⇒ <0x02> <0x00 0x05> <0x01 0x0D> <0x01> <0xFF 0xEC> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x01 0x0D> <0x00> <0xFF 0xED> <0x03>

5.15. Get LBT Settings Command (ID = 0x110)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <LBTENABLE> <CHECKSUM> <ETX>

Note:

LBTENABLE 1 Byte ; LBT Enable/Disable; 0: Disable; 1: Enable

Example:

Host ⇒ <0x02> <0x00 0x04> <0x01 0x10> <0xFF 0xEB> <0x03>

RFID ⇒ <0x02> <0x00 0x06> <0x01 0x10> <0x00> <0x00> <0xFF 0xE9> <0x03>

5.16. Set LBT Settings Command (ID = 0x111)

Host ⇒ <STX> <LENGTH> <ID> <LBTENABLE> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

LBTENABLE 1 Byte ; LBT Enable/Disable; 0: Disable; 1: Enable

Example:

Host ⇒ <0x02> <0x00 0x05> <0x01 0x11> <0x01> <0xFF 0xE8> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x01 0x11> <0x00> <0xFF 0xE9> <0x03>

5.17. Get FastID Settings Command (ID = 0x112)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <FASTIDEN> <CHECKSUM> <ETX>

Note:

FASTIDEN 1 Byte ; FastID Enable/Disable; 0: Disable; 1: Enable

Example:

Host ⇒ <0x02> <0x00 0x04> <0x01 0x12> <0xFF 0xE9> <0x03>

RFID ⇒ <0x02> <0x00 0x06> <0x01 0x12> <0x00> <0x00> <0xFF 0xE7> <0x03>

5.18. Set FastID Settings Command (ID = 0x113)

Host ⇒ <STX> <LENGTH> <ID> <FASTIDEN> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

FASTIDEN 1 Byte ; FastID Enable/Disable; 0: Disable; 1: Enable

Example:

Host ⇒ <0x02> <0x00 0x05> <0x01 0x13> <0x01> <0xFF 0xE6> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x01 0x13> <0x00> <0xFF 0xE7> <0x03>

5.19. Get TagFocus Settings Command (ID = 0x114)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <TAGFOCUSEN> <CHECKSUM> <ETX>

Note:

TAGFOCUSEN 1 Byte ; TagFocus Enable/Disable; 0: Disable; 1: Enable

Example:

Host ⇒ <0x02> <0x00 0x04> <0x01 0x14> <0xFF 0xE7> <0x03>

RFID ⇒ <0x02> <0x00 0x06> <0x01 0x14> <0x00> <0x00> <0xFF 0xE5> <0x03>

5.20. Set TagFocus Settings Command (ID = 0x115)

Host ⇒ <STX> <LENGTH> <ID> <TAGFOCUSEN> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

TAGFOCUSEN 1 Byte ; TagFocus Enable/Disable; 0: Disable; 1: Enable

Example:

Host ⇒ <0x02> <0x00 0x05> <0x01 0x15> <0x01> <0xFF 0xE4> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x01 0x15> <0x00> <0xFF 0xE5> <0x03>

5.21. Get Operation Mode Settings Command (ID = 0x116)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <OPMODE> <CHECKSUM> <ETX>

Note:

OPMODE 1 Byte ; Operation Mode: 0: Continuous; 1: Non-Continuous

Example:

Host ⇒ <0x02> <0x00 0x04> <0x01 0x16> <0xFF 0xE5> <0x03>

RFID ⇒ <0x02> <0x00 0x06> <0x01 0x16> <0x00> <0x00> <0xFF 0xE3> <0x03>

5.22. Set Operation Mode Settings Command (ID = 0x117)

Host ⇒ <STX> <LENGTH> <ID> <OPMODE> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

OPMODE 1 Byte ; Operation Mode: 0: Continuous; 1: Non-Continuous

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0x17><0x01><0xFF 0xE2><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x17><0x00><0xFF 0xE3><0x03>

5.23. Get Fixed Frequency Command (ID = 0x118)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><NUM><FREQ1>...<FREQn><CHECKSUM><ETX>

Note:

NUM 1 Byte ; Number of Fixed Frequency settings; 0: Disable

FREQ 4 Byte ; Fixed Frequency in kHz; e.g. 902750

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x18><0xFF 0xE3><0x03>

RFID ⇒ <0x02><0x00 0x0A><0x01 0x18><0x00><0x01><0x00 0x0D 0C6 0x5E>
 <0xFE 0xAB><0x03>

5.24. Set Fixed Frequency Command (ID = 0x119)

Host ⇒ <STX><LENGTH><ID><NUM><FREQ1>...<FREQn><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

NUM 1 Byte ; Number of Fixed Frequency settings; 0: Disable

FREQ 4 Byte ; Fixed Frequency in kHz; e.g. 902750

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x19><0x01><0x00 0x0D 0C6 0x5E><0xFE 0xAB><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x19><0x00><0xFF 0xE1><0x03>

5.25. Get Tx On/Off Time Command (ID = 0x11A)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><ONTIME><OFFTIME><CHECKSUM><ETX>

Note:

ONTIME 2 Byte ; Tx On time in ms.

OFFTIME 2 Byte ; Tx Off time in ms.

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x1A><0xFF 0xE1><0x03>

RFID ⇒ <0x02><0x00 0x09><0x01 0x1A><0x00><0x00 0x64><0x00 0x64>
 <0xFF 0x14><0x03>

5.26. Set Tx On/Off Time Command (ID = 0x11B)

Host ⇒ <STX><LENGTH><ID><ONTIME><OFFTIME><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ONTIME 2 Byte ; Tx On time in ms.

OFFTIME 2 Byte ; Tx Off time in ms.

Example:

Host ⇒ <0x02><0x00 0x08><0x01 0x1B><0x00 0x64><0x00 0x64><0xFF 0x14><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x1B><0x00><0xFF 0xDF><0x03>

5.27. Get Phase Data Settings Command (ID = 0x11C)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><PHASEEN><CHECKSUM><ETX>

Note:

PHASEEN 1 Byte ; Phase Data Output: 0: Disable; 1: Enable

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x1C><0xFF 0xDF><0x03>

RFID ⇒ <0x02><0x00 0x06><0x01 0x1C><0x00><0x00><0xFF 0xDD><0x03>

5.28. Set Phase Data Settings Command (ID = 0x11D)

Host ⇒ <STX><LENGTH><ID><PHASEEN><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

PHASEEN 1 Byte ; Phase Data Output: 0: Disable; 1: Enable

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0x1D><0x01><0xFF 0xDC><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x1D><0x00><0xFF 0xDD><0x03>

5.29. Get Ambient Temperature Protection Command (ID = 0x120)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><TEMPHSD><CHECKSUM><ETX>

Note:

TEMPHSD 2 Byte ; Temperature Threshold (in tenth of Celsius)

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x20><0xFF 0xDB><0x03>

RFID ⇒ <0x02><0x00 0x07><0x01 0x20><0x00><0x02 0xBC><0xFF 0x1A><0x03>

5.30. Set Ambient Temperature Protection Command (ID = 0x121)

Host ⇒ <STX><LENGTH><ID><TEMPHSD><CHECKSUM><ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

TEMPTHSD 2 Byte ; Temperature Threshold (in tenth of Celsius)

Example:

Host ⇒ <0x02> <0x00 0x06> <0x01 0x21> <0x02 0xBC> <0xFF 0x1A> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x01 0x21> <0x00> <0xFF 0xD9> <0x03>

5.31. Get PA Temperature Protection Command (ID = 0x122)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <TEMPTHSD> <CHECKSUM> <ETX>

Note:

TEMPTHSD 2 Byte ; Temperature Threshold (in tenth of Celsius)

Example:

Host ⇒ <0x02> <0x00 0x04> <0x01 0x22> <0xFF 0xD9> <0x03>

RFID ⇒ <0x02> <0x00 0x07> <0x01 0x22> <0x00> <0x02 0xBC> <0xFF 0x18> <0x03>

5.32. Set PA Temperature Protection Command (ID = 0x123)

Host ⇒ <STX> <LENGTH> <ID> <TEMPTHSD> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

TEMPTHSD 2 Byte ; Temperature Threshold (in tenth of Celsius)

Example:

Host ⇒ <0x02> <0x00 0x06> <0x01 0x23> <0x02 0xBC> <0xFF 0x18> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x01 0x23> <0x00> <0xFF 0xD7> <0x03>

5.33. Get Temperature Protection Mode Command (ID = 0x124)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <TPMODE> <CHECKSUM> <ETX>

Note:

TPMODE 1 Byte ; Temperature Protection Mode

TPMODE = 0x00 ; Stop Mode. Stop Immediately.

= 0x01 ; Non-Stop Mode. No transmission at dangerous levels, but transmit again as soon as the temperature drops.

Example:

Host ⇒ <0x02> <0x00 0x04> <0x01 0x24> <0xFF 0xD7> <0x03>

RFID ⇒ <0x02> <0x00 0x06> <0x01 0x24> <0x00> <0x00> <0xFF 0xD5> <0x03>

5.34. Set Temperature Protection Mode Command (ID = 0x125)

Host ⇒ <STX> <LENGTH> <ID> <TOMODE> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

TPMODE 1 Byte ; Temperature Protection Mode

TPMODE = 0x00 ; Stop Mode. Stop Immediately.

= 0x01 ; Non-Stop Mode. No transmission at dangerous levels, but transmit again as soon as the temperature drops.

Example:

Host ⇒ <0x02> <0x00 0x05> <0x01 0x25> <0x01> <0xFF 0xD4> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x01 0x25> <0x00> <0xFF 0xD5> <0x03>

5.35. Get Antenna Inventory Round Count Settings Command (ID = 0x126)

Host ⇒ <STX> <LENGTH> <ID> <ANTID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <ANTID> <INVRNDCNT> <CHECKSUM> <ETX>

Note:

ANTID 1 Byte ; Antenna ID; 1~4

INVRNDCNT 2 Byte ; Inventory Round Count

Example:

Host ⇒ <0x02> <0x00 0x05> <0x01 0x26> <0x01> <0xFF 0xD3> <0x03>

RFID ⇒ <0x02> <0x00 0x08> <0x01 0x26> <0x00> <0x01> <0x00 0x01> <0xFF 0xCF> <0x03>

5.36. Set Antenna Inventory Round Count Settings Command (ID = 0x127)

Host ⇒ <STX> <LENGTH> <ID> <ANTID> <INVRNDCNT> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

ANTID 1 Byte ; Antenna ID; 1~4

INVRNDCNT 2 Byte ; Inventory Round Count

Example:

Host ⇒ <0x02> <0x00 0x07> <0x01 0x27> <0x01> <0x00 0x00> <0xFF 0xD0> <0x03>

RFID ⇒ <0x02> <0x00 0x05> <0x01 0x27> <0x00> <0xFF 0xD3> <0x03>

5.37. Get Inventory Data with Customized Format Settings Command (ID = 0x128)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>

RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <PID1>...<PIDn> <CHECKSUM> <ETX>

Note:

PID	= 0x00	; Display EPC in Data Reply if possible
	= 0x01	; Display PC in Data Reply if possible
	= 0x02	; Display RSSI in Data Reply if possible
	= 0x03	; Display Antenna ID in Data Reply if possible
	= 0x04	; Display TID in Data Reply if possible
	= 0x05	; Display User specified memory in Data Reply if possible
	= 0x06	; Display Phase in Data Reply if possible
	= 0x07	; Display Frequency in Data Reply if possible
	= 0x08	; Display Timestamp in Data Reply if possible

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x28><0xFF 0xD3><0x03>
 RFID ⇒ <0x02><0x00 0x09><0x01 0x28><0x00><0x00><0x00 0x01 0x04 0x05>
 <0xFF 0xC4><0x03>

5.38. Set Inventory Data with Customized Format Settings Command (ID = 0x129)

Host ⇒ <STX><LENGTH><ID><PID1>...<PIDn><CHECKSUM><ETX>
 RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

PID	= 0x00	; Display EPC in Data Reply if possible
	= 0x01	; Display PC in Data Reply if possible
	= 0x02	; Display RSSI in Data Reply if possible
	= 0x03	; Display Antenna ID in Data Reply if possible
	= 0x04	; Display TID in Data Reply if possible
	= 0x05	; Display User specified memory in Data Reply if possible
	= 0x06	; Display Phase in Data Reply if possible
	= 0x07	; Display Frequency in Data Reply if possible
	= 0x08	; Display Timestamp in Data Reply if possible

Giving an empty PID means disabling the customized format.

Example:

Host ⇒ <0x02><0x00 0x06><0x01 0x29><0x00 0x08><0xFF 0xC8><0x03>
 RFID ⇒ <0x02><0x00 0x05><0x01 0x29><0x00><0xFF 0xD1><0x03>

5.39. Get System Time Command (ID = 0x12A)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>
 RFID ⇒ <STX><LENGTH><ID><STATUS><YEAR><MON><DAY><HOUR>
 <MIN><SEC><CHECKSUM><ETX>

Note:

YEAR	1 Byte	; Year (from 2000, e.g. 1 for 2001)
------	--------	-------------------------------------

MON	1 Byte	; Month
DAY	1 Byte	; Day
HOUR	1 Byte	; Hour
MIN	1 Byte	; Minute
SEC	1 Byte	; Second

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0x26><0x01><0xFF 0xD3><0x03>

RFID ⇒ <0x02><0x00 0x08><0x01 0x26><0x00><0x01><0x00 0x01><0xFF 0xCF><0x03>

5.40. Set System Time Command (ID = 0x12B)

Host ⇒ <STX><LENGTH><ID><YEAR><MON><DAY><HOUR>
<MIN><SEC><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

YEAR	1 Byte	; Year (from 2000, e.g. 1 for 2001)
MON	1 Byte	; Month
DAY	1 Byte	; Day
HOUR	1 Byte	; Hour
MIN	1 Byte	; Minute
SEC	1 Byte	; Second

Example:

Host ⇒ <0x02><0x00 0x07><0x01 0x27><0x01><0x00 0x00><0xFF 0xD0><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x27><0x00><0xFF 0xD3><0x03>

5.41. Get Power Saving Mode Command (ID = 0x130)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><PSMODE><CHECKSUM><ETX>

Note:

PSMODE 1 Byte ; Power Saving Mode

PSMODE = 0x00 ; Power-Saving Mode.

= 0x01 ; Power-On Mode. Always Power-On.

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x30><0xFF 0xCB><0x03>

RFID ⇒ <0x02><0x00 0x06><0x01 0x30><0x00><0x00><0xFF 0xC9><0x03>

5.42. Set Power Saving Mode Command (ID = 0x131)

Host ⇒ <STX><LENGTH><ID><PSMODE><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

PSMODE 1 Byte ; Power Saving Mode

PSMODE = 0x00 ; Power-Saving Mode.

 = 0x01 ; Power-On Mode. Always Power-On.

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0x31><0x01><0xFF 0xC8><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x31><0x00><0xFF 0xC9><0x03>

5.43. Get Power Saving Timeout Command (ID = 0x132)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><PSTIME><CHECKSUM><ETX>

Note:

PSTIME 2 Byte ; Power Saving Timeout (in ms): Idle time to Sleep

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x32><0xFF 0xC9><0x03>

RFID ⇒ <0x02><0x00 0x07><0x01 0x32><0x00><0x0B 0xB8><0xFF 0x03><0x03>; 3000 ms

5.44. Set Power Saving Timeout Command (ID = 0x133)

Host ⇒ <STX><LENGTH><ID><PSTIME><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

PSTIME 2 Byte ; Power Saving Timeout (in ms): Idle time to Sleep

Example:

Host ⇒ <0x02><0x00 0x06><0x01 0x33><0x0B 0xB8><0xFF 0x03><0x03>; 3000 ms

RFID ⇒ <0x02><0x00 0x05><0x01 0x33><0x00><0xFF 0xC7><0x03>

5.45. Get UART Default Baud Rate Command (ID = 0x150)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><BAUDRATE><CHECKSUM><ETX>

Note:

BAUDRATE 4 Byte ; UART Baud Rate; 115200/460800/921600

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x50><0xFF 0xAB><0x03>
RFID ⇒ <0x02><0x00 0x09><0x01 0x50><0x00><0x00 0x01 0xC2 0x00><0xFE 0xE3><0x03>

5.46. Set UART Default Baud Rate Command (ID = 0x151)

Host ⇒ <STX><LENGTH><ID><BAUDRATE><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note: New baud rate will take effect on the next device reboot.

BAUDRATE 4 Byte ; UART Baud Rate; 115200/460800/921600

Example:

Host ⇒ <0x02><0x00 0x08><0x01 0x51><0x00 0x0E 0x10 0x00><0xFF 0x88><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x51><0x00><0xFF 0xA9><0x03>

5.47. Get GPIO Pins Configuration Command (ID = 0x180)

Host ⇒ <STX><LENGTH><ID><PINNUM><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><PINNUM><PINDIR><CHECKSUM><ETX>

Note:

PINNUM 1 Byte ; GPIO Pin Number

PINDIR 1 Byte ; GPIO Pin Direction; 0 for input; 1 for output; 2 for Alt-Func if avail

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0x80><0x02><0xFF 0x78><0x03>

RFID ⇒ <0x02><0x00 0x07><0x01 0x80><0x00><0x02><0x00><0xFF 0x76><0x03>

5.48. Set GPIO Pins Configuration Command (ID = 0x181)

Host ⇒ <STX><LENGTH><ID><PINNUM><PINDIR><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

PINNUM 1 Byte ; GPIO Pin Number

PINDIR 1 Byte ; GPIO Pin Direction; 0 for input; 1 for output; 2 for Alt-Func if avail

Example:

Host ⇒ <0x02><0x00 0x06><0x01 0x81><0x02><0x01><0xFF 0x75><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0x81><0x00><0xFF 0x79><0x03>

5.49. Get GPIO Pins State Command (ID = 0x182)

Host ⇒ <STX><LENGTH><ID><PINNUM><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><PINNUM><PINSTATE><CHECKSUM><ETX>

Note:

PINNUM 1 Byte ; GPIO Pin Number

PINSTATE 1 Byte ; GPIO Pin State; 0 for low; 1 for high

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0x82><0x02><0xFF 0x76><0x03>
RFID ⇒ <0x02><0x00 0x07><0x01 0x82><0x00><0x02><0x00><0xFF 0x74><0x03>

5.50. Set GPIO Pins State Command (ID = 0x183)

Host ⇒ <STX><LENGTH><ID><PINNUM><PINSTATE><CHECKSUM><ETX>
RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

PINNUM 1 Byte ; GPIO Pin Number
PINSTATE 1 Byte ; GPIO Pin State; 0 for low; 1 for high

Example:

Host ⇒ <0x02><0x00 0x06><0x01 0x83><0x02><0x01><0xFF 0x73><0x03>
RFID ⇒ <0x02><0x00 0x05><0x01 0x83><0x00><0xFF 0x77><0x03>

5.51. Get LBT Configure Settings Command (ID = 0x190)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>
RFID ⇒ <STX><LENGTH><ID><STATUS><RSSITHSD><CHECKSUM><ETX>

Note:

RSSITHSD 2 Byte ; LBT RSSI Threshold in cdBm; e.g. -7400 for -74 dBm.

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0x90><0xFF 0x6B><0x03>
RFID ⇒ <0x02><0x00 0x07><0x01 0x90><0x00><0xE3 0x18><0xFE 0x6D><0x03>

5.52. Set LBT Configure Settings Command (ID = 0x191)

Host ⇒ <STX><LENGTH><ID><RSSITHSD><CHECKSUM><ETX>
RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

RSSITHSD 2 Byte ; LBT RSSI Threshold in cdBm; e.g. -7400 for -74 dBm.

Example:

Host ⇒ <0x02><0x00 0x06><0x01 0x91><0xE3 0x18><0xFE 0x6D><0x03>
RFID ⇒ <0x02><0x00 0x05><0x01 0x91><0x00><0xFF 0x69><0x03>

5.53. Get Antenna Switching Mode for Single Antenna Command (ID = 0x1A0)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>
RFID ⇒ <STX><LENGTH><ID><STATUS><ASMODE><CHECKSUM><ETX>

Note:

ASMODE 1 Byte ; Antenna Switching Mode for Single Antenna

ASMODE = 0x00 ; Stop Mode. Stop RF output during antenna switching.
= 0x01 ; Non-Stop Mode. Keep RF output during antenna switching.

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0xA0><0xFF 0x5B><0x03>
RFID ⇒ <0x02><0x00 0x06><0x01 0xA0><0x00><0x00><0xFF 0x59><0x03>

5.54. Set Antenna Switching Mode for Single Antenna Command (ID = 0x1A1)

Host ⇒ <STX><LENGTH><ID><ASMODE><CHECKSUM><ETX>
RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ASMODE 1 Byte ; Antenna Switching Mode for Single Antenna

ASMODE = 0x00 ; Stop Mode. Stop RF output during antenna switching.
= 0x01 ; Non-Stop Mode. Keep RF output during antenna switching.

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0xA1><0x01><0xFF 0x58><0x03>
RFID ⇒ <0x02><0x00 0x05><0x01 0xA1><0x00><0xFF 0x59><0x03>

5.55. Get Frequency Switching Mode for Single Frequency Command (ID = 0x1A2)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>
RFID ⇒ <STX><LENGTH><ID><STATUS><FSMODE><CHECKSUM><ETX>

Note:

FSMODE 1 Byte ; Frequency Switching Mode for Single Frequency

FSMODE = 0x00 ; Stop Mode. Stop RF output during frequency switching.
= 0x01 ; Non-Stop Mode. Keep RF output during frequency switching.

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0xA2><0xFF 0x59><0x03>
RFID ⇒ <0x02><0x00 0x06><0x01 0xA2><0x00><0x00><0xFF 0x57><0x03>

5.56. Set Frequency Switching Mode for Single Frequency Command (ID = 0x1A3)

Host ⇒ <STX><LENGTH><ID><FSMODE><CHECKSUM><ETX>
RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

FSMODE 1 Byte ; Frequency Switching Mode for Single Frequency

FSMODE = 0x00 ; Stop Mode. Stop RF output during frequency switching.
= 0x01 ; Non-Stop Mode. Keep RF output during frequency switching.

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0xA3><0x01><0xFF 0x56><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0xA3><0x00><0xFF 0x57><0x03>

5.57. Get Antenna Detection Settings Command (ID = 0x1A4)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><ANTDET><CHECKSUM><ETX>

Note:

ANTDET 1 Byte ; Antenna Detection Settings

ANTDET = 0x00 ; Disable Antenna Detection.

 = 0x01 ; Enable Antenna Detection.

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0xA4><0xFF 0x57><0x03>

RFID ⇒ <0x02><0x00 0x06><0x01 0xA4><0x00><0x00><0xFF 0x55><0x03>

5.58. Set Antenna Detection Settings Command (ID = 0x1A5)

Host ⇒ <STX><LENGTH><ID><ANTDET><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ANTDET 1 Byte ; Antenna Detection Settings

ANTDET = 0x00 ; Disable Antenna Detection.

 = 0x01 ; Enable Antenna Detection.

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0xA5><0x01><0xFF 0x54><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0xA5><0x00><0xFF 0x55><0x03>

5.59. Get Configuration Storage Mode Command (ID = 0x1F0)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><STMODE><CHECKSUM><ETX>

Note:

STMODE 1 Byte ; RFID Configuration Storage Mode

STMODE = 0x00 ; Save-to RAM/Flash Mode.

 = 0x01 ; Save-to RAM only Mode.

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0xF0><0xFF 0x0B><0x03>

RFID ⇒ <0x02><0x00 0x06><0x01 0xF0><0x00><0x00><0xFF 0x09><0x03>

5.60. Set Configuration Storage Mode Command (ID = 0x1F1)

Host ⇒ <STX><LENGTH><ID><STMODE><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

STMODE 1 Byte ; RFID Configuration Storage Mode

STMODE = 0x00 ; Save-to RAM/Flash Mode.

 = 0x01 ; Save-to RAM only Mode.

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0xF1><0x01><0xFF 0x08><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0xF1><0x00><0xFF 0x09><0x03>

5.61. Get Error Suppression Mode Command (ID = 0x1F2)

Host ⇒ <STX><LENGTH><ID><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><ESMODE><CHECKSUM><ETX>

Note:

ESMODE 1 Byte ; Error Suppression Mode.

ESMODE = 0x00 ; Report all Ex10 Errors.

 = 0x01 ; Suppress all Ex10 Errors.

Example:

Host ⇒ <0x02><0x00 0x04><0x01 0xF2><0xFF 0x09><0x03>

RFID ⇒ <0x02><0x00 0x06><0x01 0xF2><0x00><0x00><0xFF 0x07><0x03>

5.62. Set Error Suppression Mode Command (ID = 0x1F3)

Host ⇒ <STX><LENGTH><ID><ESMODE><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note:

ESMODE 1 Byte ; Error Suppression Mode.

ESMODE = 0x00 ; Report all Ex10 Errors.

 = 0x01 ; Suppress all Ex10 Errors.

Example:

Host ⇒ <0x02><0x00 0x05><0x01 0xF3><0x01><0xFF 0x06><0x03>

RFID ⇒ <0x02><0x00 0x05><0x01 0xF3><0x00><0xFF 0x07><0x03>

6. RFID Extended Operation

6.1. CW On Command (ID = 0xF00)

Host ⇒ <STX> <LENGTH> <ID> <ANTID> <PWR> <FREQ> <CHECKSUM> <ETX>
RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

ANTID	1 Byte	; Testing Antenna ID; 1~4
PWR	2 Byte	; Testing Antenna Power in cdBm
FREQ	4 Byte	; Testing Frequency in kHz; e.g. 902750

Example:

Host ⇒ <0x02> <0x00 0x0B> <0x0F 0x00> <0x01> <0x0B 0xB8> <0x00 0x0D 0xC6 0x5E>
 <0xFD 0xF1> <0x03>
RFID ⇒ <0x02> <0x00 0x05> <0x0F 0x00> <0x00> <0xFF 0xEC> <0x03>

6.2. CW Off Command (ID = 0xF01)

Host ⇒ <STX> <LENGTH> <ID> <CHECKSUM> <ETX>
RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Example:

Host ⇒ <0x02> <0x00 0x04> <0x0F 0x01> <0xFF 0xEC> <0x03>
RFID ⇒ <0x02> <0x00 0x05> <0x0F 0x01> <0x00> <0xFF 0xEB> <0x03>

6.3. Tx Random Data Command (ID = 0xF02)

Host ⇒ <STX> <LENGTH> <ID> <ANTID> <PWR> <FREQ> <DURATION> <CHECKSUM> <ETX>
RFID ⇒ <STX> <LENGTH> <ID> <STATUS> <CHECKSUM> <ETX>

Note:

ANTID	1 Byte	; Testing Antenna ID; 1~4
PWR	2 Byte	; Testing Antenna Power in cdBm
FREQ	4 Byte	; Testing Frequency in kHz; e.g. 902750
DURATION	4 Byte	; Testing Auto Off duration in ms.

Example:

Host ⇒ <0x02> <0x00 0x0F> <0x0F 0x02> <0x01> <0x0B 0xB8> <0x00 0x0D 0xC6 0x5E>
 <0x00 0x00 0x0B 0xB8> <0xFD 0x28> <0x03>
RFID ⇒ <0x02> <0x00 0x05> <0x0F 0x02> <0x00> <0xFF 0xEA> <0x03>

7. Event Notification Commands

7.1. Event Notification Enable/Disable Command (ID = 0x90)

Host ⇒ <STX><LENGTH><ID><EVENTID><DATA><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note: EVENTID 1 Byte ; Event ID

DATA 1 Byte ; Enable:1 ; Disable:0

EVENTID = 0xE0 ; RFID Antenna Begin/End.

= 0xE1 ; RFID LBT Status.

Example:

Host ⇒ <0x02><0x00 0x06><0x00 0x90><0xE0><0x01><0xFE 0x89><0x03>

RFID ⇒ <0x02><0x00 0x05><0x00 0x90><0x00><0xFF 0x68><0x03>

7.2. Event Notification Command - Antenna Status (ID = 0xE0)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <ANTID><ANTSTATE>

ANTID 1 Byte ; Antenna ID; 1~4

ANTSTATE 1 Byte ; Begin: 0x00; End: 0x01

Example:

RFID ⇒ <0x02><0x00 0x06><0x00 0xE0><0x01><0x00><0xFF 0x19><0x03>

7.3. Event Notification Command - LBT Status (ID = 0xE1)

RFID ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

Note: DATA = <INTERF><RSSI><FREQ>

INTERF 1 Byte ; Interferer status: 0: No Interferer; 1: Interferer present.

RSSI 2 Byte ; RSSI value

FREQ 4 Byte ; Testing Frequency in kHz; e.g. 902750

Example:

RFID ⇒ <0x02><0x00 0x0B><0x00 0xE1><0x00><0xE1 0x10><0x00 0x0D 0xE7 0x92>
<0xFC 0x9D><0x03>

8. Firmware Update Commands

8.1. Start File Transfer (Op-ID = 0x6101)

Host ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note: DATA = "File Name" + "\0" + "File Size" + "\0"

This operation is only for File update mode.

Example:

Host ⇒ <0x02><0x00 0x15><0x61 0x01>
<0x53><0x4C><0x32><0x32><0x30><0x46><0x57><0x2E>
<0x62><0x69><0x6E><0x00>
<0x34><0x30><0x39><0x36><0x00>
<0xFB 0x7F><0x03>
; DATA = "SL220FW.bin\04096\0"

RFID ⇒ <0x02><0x00 0x05><0x61 0x01><0x00><0xFF 0xF9><0x03>

8.2. Send File Data (Op-ID = 0x6102)

Host ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note: DATA = Firmware binary packet (Multi-Packets; Max Size: 1024 Bytes for RM300P)

This operation is only for File Update Mode.

Example:

Host ⇒ <0x02><0x00 0x15><0x61 0x02>
<0x20><0x01><0x07><0xE8><0x80><0x02><0x03><0xA1>
<0x08><0x02><0x3D><0x71><0x08><0x02><0x3C><0x6F>
<0x08>
<0xFB 0xDD><0x03>

RFID ⇒ <0x02><0x00 0x05><0x61 0x02><0x00><0xFF 0xF8><0x03>

8.3. Perform Firmware Update (Op-ID = 0x6103)

Host ⇒ <STX><LENGTH><ID><DATA><CHECKSUM><ETX>

RFID ⇒ <STX><LENGTH><ID><STATUS><CHECKSUM><ETX>

Note: DATA = 1 ; File Update Mode (For RM300P)

= 2 ; Y-Modem Update Mode

Example:

Host ⇒ <0x02><0x00 0x05><0x61 0x03><0x01><0xFF 0x96><0x03>

RFID ⇒ <0x02><0x00 0x05><0x61 0x03><0x00><0xFF 0x97><0x03>